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Python - Introduction

Review Questions

1. The _____ types in python are the built-in types for which we cannot change their content once created.
2. `int` and `float` data types are Immutable type. State True / False.
3. In python `bool` data type is a subclass of _____
4. The `bool` data types values `True` and `False` are represented as _____ and _____.
5. _____ is a mutable data type.
6. Write a statement to assign the value 10 and 20 to `a` and `b` variables.

7. Write a statement to assign the value 50 to `a`, `b` and `c` variables.

8. `x = "65"`. Write a statement to convert `x` to `int` type. _____.

Exercises

- Write the expected output, or errors if any, for each of the following programs in the box provided below each program.
- Then execute the programs and check your answers.
- Also answer the questions given below.

Program 1

```
1 x = 10
2 y = 20
3 print(x)
4 print(y)
```

Program 2

```
1 a = 40
2 b = 50
3 c = a + b
4 print(c)
```

Program 3

```
1 x = 30
2 y = 30
3 print(x == y)
```

Q1: What will be the output if the value of **x** is greater than **y**.

Q2: What will be the output if the value of **x** is less than **y**.

Program 4

```
1 a = 45
2 b = 65
3 a, b = b, a
4 print(a, b)
```

Program 5

```
1 a = 37
2 b = a / 3
3 print(b)
```

Q1: What will be the output, if line 2 is replaced by **b = a // 3**?

Program 6

```
1 n = 5
2 r = n * 5
3 print(r)
```

Q1: What will be the output, if line 2 is replaced with `r = n ** 5`?

Program 7

```
1 a = 242
2 print(a % 10 == a // 100)
```

Q1: What will be the output, if line 2 is replaced with `print(a % 10 == a / 100)`?

Program 8

```
1 m1 = 65
2 m2 = 65
3 m3 = 65
4 print(m1 >= 65 and m2 >= 65 and m3 >= 65)
```

Q1: What will be the output if `m2 = 60`?

Q2: What will be the output if values of `m1`, `m2`, and `m3` are 50?

Q3: What will be the output if in line 4 `and` is replaced by `or`?

Q4: What will be the output if in line 4, `and` is replaced by `or`, as well as values 50, 40 and 70 for `m1`, `m2` and `m3`?

Additional Exercises

1. Write a program to calculate total marks and average of three subject marks.
2. Write a program that prints the power of one number to the other.
3. Write a program to print the sum of all individual digits of any three digits number.
4. Write a program which prints **True** if the number is Armstrong (Only Three Digits Numbers).

Python - If...Else

Exercises

- Write the expected output, or errors if any, for each of the following programs in the box provided below each program.
- Then execute the programs and check your answers.
- Also answer the questions given below.

Program 1

```
1 num = int(input('Enter The Number: '))
2 if num % 2 == 0:
3     print('Even Number')
4 else:
5     print('Odd Number')
```

Q1: What will be the output for a value 33?

Q2: What will be the output for a value 56?

Program 2

```
1 a = int(input('Enter the First Number: '))
2 b = int(input('Enter the Second Nummber: '))
3 c = int(input('Enter the Third Number: '))
4 if a > b and a > c:
5     print('A is Greater')
6 elif b > c:
7     print('B is Greater')
8 else:
9     print('C is Greater')
```

Q1: What will be the output if the value of **a**, **b** and **c** are 10, 20 and 30?

Q2: What will be the output if the value of **a**, **b** and **c** are equal to 55?

Q3: Modify the if statement without using the **and** operator

Program 3

```
1 num = 1
2 limit = 20
3 sum_num = 0
4 while num <= limit:
5     if num % 3 == 0:
6         sum_num += num
7     num += 1
8 print(sum_num)
```

Program 4

```
1 num = 0
2 limit = 20
3 sum_d = 0
4 while num <= limit:
5     num += 1
6     if num % 2 == 0:
7         continue
8     sum_d += num
9 print(sum_d)
```

Program 5

```
1 nums = [22, 33, 44, 55, 66, 77, 88, 99]
2 n = int(input('Enter The Number: '))
3 if n in nums:
4     print('Element Exist')
5 else:
6     print('Element Not Exist')
```

Q1: What will be the output if the input value is 33?

Q2: What will be the output if the input value is 100?

Additional Exercises

1. Given a year, check the year is a leap year.
2. Given a month, find the number of days in that month.
3. Write a program to print the greatest value among three values.
4. Write a program to check the given number is a palindrome number.
5. Write a program to print the collatz sequence for a given number.
Note: To get a Collatz sequence from a number, if it's even, divide it by two, and if it's odd, multiply it by three and add one. Continue the operation on the result of the previous operation until the number becomes 1.
6. Write a program to find the product of two given numbers using russian multiplication.
7. Write a program to check the given number is a perfect number.
8. Write a program to check the given number is a prime number.
9. Write a program to print the sum of all even digits in a given number.

Python - While Loop

Exercises

- Write the expected output, or errors if any, for each of the following programs in the box provided below each program.
- Then execute the programs and check your answers.
- Also answer the questions given below.

Program 1

```
1 n = 1
2 while n <= 10:
3     print(n)
4     n += 1
```

Program 2

```
1 n = 2
2 n2 = int(input('Enter The Number: '))
3 while n <= n2:
4     print(n)
5     n += 2
```

Program 3

```
1 num = 20
2 while num >= 1:
3     print(num)
4     num -= 1
```

Program 4

```
1 num = int(input('Enter The Number: '))
2 x = 0
3 while num > 0:
4     num //= 10
5     x += 1
6 print(x)
```

Explain your understanding on the above code.

Program 5

```
1 j = 2
2 while j <= 10:
3     print(j * j)
4     j += 1
```

Program 6

```
1 i = 1
2 while i <= 5:
3     print('* ' * i)
4     i += 1
```

Program 7

```
1 i = 1
2 while i <= 5:
3     j = 1
4     while j <= 5:
5         print('* ', end = ' ')
6         j += 1
7     print()
8     i += 1
```

Program 8

```

1 num = int(input('Enter The Number: '))
2 i = 0
3 while num > 0:
4     x = num % 10
5     i = i + (x * x)
6     num //= 10
7 print(i)

```

Write the output of the above code and explain.

Program 9

```

1 num = int(input('Enter The Number: '))
2 res = 0
3 while num > 0:
4     r = num % 10
5     res = res + (r ** r)
6     num //= 10
7 print(res)

```

Write the output of the above code and explain.

Additional Exercises

1. Write a program to print the sum of numbers with in a given range.
2. Write a program to accepts three numbers (Start Value, End Value and Increment Value) as input and print the numbers from start value to end value incrementing by the given value.
3. Write a program that prints first 10 numbers in a fibonacci series.

4. Write programs to print the following patterns for an input value 5.

(a) Pattern - 1

```
*
* *
* * *
* * * *
* * * * *
```

(b) Pattern - 2

```
5 4 3 2 1
4 3 2 1
3 2 1
2 1
1
```

(c) Pattern - 3

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

(d) Pattern - 4

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

(e) Pattern - 5

```
      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * *
* * * * *
```

(f) Pattern - 6

* * * * *

(f) Pattern - 6

```
      *
    * + *
  * + * + *
* + * + * + *
* + * + * + * + *
```

Python - Strings

Review Questions

1. The string value can be enclosed in _____ and _____ quotes.
2. Write a statement to store *I don't understand!* in a variable **message**?

3. Multi line string has to be enclosed in _____ quotes.
4. `str1 = "Good "` `str2 = "Morning"`
Write the statement to concatenate *str1* and *str2*._____.
5. `a = 101` and `b = 102`
Write a statement which prints '101102' as output _____.
6. The return type of the *split()* function is _____.
7. The function used to convert *number* to *string* is _____.
8. Write a statement to check whether string *s1* contains *s2* _____.

Exercises

- Write the expected output, or errors if any, for each of the following programs in the box provided below each program.
- Then execute the programs and check your answers.
- Also answer the questions given below.

Program 1

```
1 str1 = 'Hello World! '
2 print(len(str1))
```

Program 2

```
1 a = 10
2 b = 20
3 c = str(a) + str(b)
4 print(c)
```

Program 3

```
1 Name1 = 'Vinod'
2 Name2 = 'Kumar'
3 fullname = Name1 + ' ' + Name2
4 print(fullname)
```

Q1: What will be the output if line 4 is replaced with **print(len(fullname))**?

Program 4

```
1 wish = 'WELCOME'
2 print(wish[::-1])
```

Program 5

```
1 s = 'TalentSprint'
2 print(s[:6])
```

Q1: What will be the output, if line 2 is replaced by **print(s[6])**?

Q2: What will be the output, if line 2 is replaced by **print(s[6:])**?

Program 6

```
1 str1 = 'madam'
2 print(str1 == str1[::-1])
```

Q1: Explain your understanding on the above code.

Q2: What will be the output if line 1 is replaced with `str1 = "Madam"`?

Program 7

```
1 str1 = 'hello'
2 new_str = str1.replace('l','e')
3 print(new_str)
```

Q1: What will be the output if line 1 is replaced with `str1 = "HELLO"`?

Program 8

```
1 colors = 'red , blue , green , white , yellow , purple '
2 list_colors = colors.split(',')
3 print(list_colors[1])
```

Q1: What will be the output if line 4 is replaced by `print(len(list_colors))`?

Q2: What will be the output if line 4 is replaced by `print(list_colors[-1])`?

Q3: What will be the output if line 4 is replaced by `print(list_colors)`?

Program 9

```
1 str1 = ' Hello Good Morning '
2 print(len(str1))
```

Q1: What will be the output if line 2 is replaced with `print(len(str1.strip()))`?

Additional Exercises

1. Write a program that accepts the month in the form of string and display the number of days in that month.
2. Write a program to print the character frequency of the given string.
3. Write a program to calculate the score for a given string.

Score: The sum of alphabetical value of each character in the given string.

Alphabetical Value: The position of the alphabet in an alphabetical order. Ex: A - 1, B - 2, .. Z - 26

Input: TEST

Output: $20 + 5 + 19 + 20 = 64$

4. Given the date in the form of DD-MM-YYYY, Write a program to find the day of the year.

Input: 21-02-2013

Output: 52

Python - For Loop

Review Questions

1. `list(range(5))` will print _____
2. `list(range(1, 10))` will print _____
3. `list(range(1, 25, 2))` will print _____
4. `list(range(10, 0, -1))` will print _____
5. `list(range(5, 100, 10))` will print _____
6. `sum(range(1, 15, 2))` will print _____
7. `list(range(5, 50, 5))[-1]` will print _____
8. `list(range(2, 25, 2))[5]` will print _____

Exercises

- Write the expected output, or errors if any, for each of the following programs in the box provided below each program.
- Then execute the programs and check your answers.
- Also answer the questions given below.

Program 1

```
1 str1 = 'HelloWorld'
2 print(list(str1))
```

Program 2

```
1 for i in range(1, 11):
2     print(i * i)
```

Program 3

```
1 new_list = []
2 str1 = 'Hello'
3 str2 = 'World'
4 if len(str1) == len(str2):
5     for indx in range(len(str1)):
6         new_list.append(str1[indx])
7         new_list.append(str2[indx])
8     print(new_list)
9 else:
10    print('Both String Must Have Same Length')
```

Q1: What will be the output if value of str2 = "Welcome"?

Program 4

```
1 s = 0
2 for num in range(2, 25, 2):
3     s += num
4 print(s)
```

Program 5

```
1 s = 'WELCOME'
2 for i in range(0, len(s), 2):
3     print(s[i], end = ' ')
```

Q1: What will be the output if line 2, is replaced by **for i in range(len(s)):**?

Q2: What will be the output if line 2, is replaced by **for i in range(len(s), 2):**?

Q3: What will be the output if line 2, is replaced by **for i in range(0, len(s)):**?

Q4: What will be the output if line 2, is replaced by **for i in range(len(s) -1, -1, -1):**?

Additional Exercises

1. Write a program to print all three digits palindrome numebbers?
2. Write a program to print all prime number with in a given range?

Python - List

Review Questions

1. Lists are used to group _____
2. Lists items are enclosed within _____ brackets and separated by _____.
3. An empty list is denoted by _____.
4. List can contain items of different data types. (Yes / No)
5. Write a statement which stores first five odd numbers in a list with the name `OddList`. _____
6. The elements of a list can be accessed by its _____.

Exercises

- Write the expected output, or errors if any, for each of the following programs in the box provided below each program.
- Then execute the programs and check your answers.
- Also answer the questions given below.

Program 1

```
1 list1 = [5, 10, 15, 20, 25]
2 print(len(list1))
```

Program 2

```
1 list1 = [2, 4, 6, 8, 10]
2 print(list1[0])
```

Program 3

```
1 list1 = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
2 print(list1[len(list1) - 1])
```

Program 4

```
1 list1 = [1, 3, 5, 7, 9]
2 print(list1[-1])
```

Program 5

```
1 list1 = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
2 print(sum(list1))
```

Program 6

```
1 list1 = [22, 33, 44, 55, 66, 77]
2 print(list1[len(list1) - 1] == list1[-1])
```

Program 7

```
1 data = ["H", "E", "L", "L", "O"]
2 data[1] = "A"
3 print(data)
```

Program 8

```
1 nums = [11, 12, 13, 14, 15, 16, 17, 18, 19, 20]
2 print(nums[1:5])
```

Program 9

```
1 num1 = [1, 2, 3, 4, 5]
2 num2 = [6, 7, 8, 9, 10]
3 num1 = num1 + num2
4 print(num1)
```

Program 10

```
1 nums = [12, 23, 45, 56, 67, 78]
```

Write a statement to add an element with value **88** after **45**.

Program 11

```
1 nums = [22, 33, 44, 55, 66, 77, 88, 99]
2 nums.clear()
3 print(nums)
```

Program 12

```
1 nums = [22, 33, 44, 55, 33, 44, 55, 66]
2 nums.remove(33)
3 print(nums)
```

Additional Exercises

1. Write a program which print **True** if marks secured by the given student is more than 250 otherwise print **False**.
2. Write a program which displays the number of days in a given month.

Python - Dictionaries

Review Questions

1. Dictionary is a mapping between a _____ and _____.
2. _____ function is used to create a new dictionary with no items.
3. Dictionaries define _____ relationship between keys and values.
4. Dictionaries do not normally have any notion of order. (True or False)
5. Write a statement to create an empty dictionary _____.
6. The symbol used to separate the **key** and **value** in a dictionary is _____.
7. The **values** of a dictionary can be of any data type. (True / False)
8. _____ are unique within a dictionary.
9. The function used to find the size of a dictionary is _____.
10. The function used to clear all the items in a dictionary is _____.
11. The items in a dictionary can be accessed with an index. State True / False.
12. The _____ function returns a list of all the values available in a dictionary.
13. Write a statement to delete a dictionary '**d1**' _____.
14. The operator used to check whether a particular **key** exist in a dictionary is _____.

Code Reading Exercises

- Write the expected output, or errors if any, for each of the following programs in the box provided below each program.
- Then execute the programs and check your answers.

- Also answer the questions given below.

Program 1:

```

1      def fun1():
2          d1 = {}
3          for i in range(1, n + 1):
4              d1[i] = i * i
5          return d1
6
7      print(fun1(5))

```

Q1: What will be the output if line 4 is replaced with `d1[i] = i ** i`?

Program 2:

```

1      d1 = {'1': 'One', '2': 'Two', '3': 'Three',
2           '4': 'Four', '5': 'Five', '6': 'Six',
3           '7': 'Seven', '8': 'Eight',
4           '9': 'Nine', '0': 'Zero'}
5      x = input('Enter The Number') # Let x be 2576
6      for let in x:
7          print(d1[let], end = ' ');

```

Q1: What will be the output if the value of `x` is 23234?

Q2: What will be the output if the value of `x` is 304?

Program 3:

```

1      def fun2(val):
2          d1 = {1: 'One', 2: 'Two', 3: 'Three',
3               4: 'Four', 5: 'Five', 6: 'Six',
4               7: 'Seven', 8: 'Eight', 9: 'Nine'}
5          return d1[len(val)] + ' Digit Number'
6
7      print(fun2('45254'))

```

Q1: What will be the output if argument value is 352?

Q2: What will be the output if argument value is 5427?

Q3: What will be the output if argument value is 5637835?

Program 4:

```

1      def XYZ(mon):
2          d1 = {'JAN': 31, 'FEB': 28, 'MAR': 31,
3               'APR': 30, 'MAY': 31, 'JUN': 30,
4               'JUL': 31, 'AUG': 31, 'SEP': 30,
5               'OCT': 31, 'NOV': 30, 'DEC': 31}
6          if mon in d1:
7              return d1[mon]
8          return -1
9
10     print(XYZ('FEB'))

```

Q1: What will be the output if the argument value is "MAY"?

Q2: What will be the output if the argument value is "APRIL"?

Q3: What will be the output if the argument value is "January"?

Q4: What will be the output if the argument value is "may"?

Program 5:

```

1      def char_frequency(string):
2          dcount = {}
3          for ch in string:
4              if ch is not ' ':
5                  if ch in dcount.keys():
6                      dcount[ch] += 1
7                  else:
8                      dcount[ch] = 1
9          return dcount
10
11     print(char_frequency('TRY IT BY YOURSELF'))

```

Program 6:

```

1      def count(s):
2          dcount = {}
3          s = s.lower()
4          for ch in s:
5              if ch in 'aeiou':
6                  if ch in dcount.keys():
7                      dcount[ch] += 1
8              else:
9                  dcount[ch] = 1
10         return len(dcount)
11
12     print(count('EDUCATION'))

```

Q1: What will be the output if the argument value is “REVOLUTIONARIES”?

Q2: What will be the output if the argument value is “CRYPT”?

Coding Exercises

1. Write a program to display all anagram strings.

Note: An anagram of a word, is a new word got by rearranging the letters of a given word. SILENT and LISTEN are anagrams.

Input : “pest step ball palm heat listen post enlist silent stop lamp”

Output :

```

pest step
listen enlist silent
post stop

```

2. Write a program to implement Rock Paper Scissors game. The winner of Rock Paper Scissors is determined as follows:

Rock defeats (breaks) Scissors.

Scissors defeats (cuts) Paper.

Paper defeats (covers) Rock.

If both players choose the same item, the game is a tie and they play again.

Note: For further information please visit: <http://en.wikipedia.org/wiki/Rock-paper-scissors>